Questions

Q1.

(i) Figure 17 shows the output from a battery.

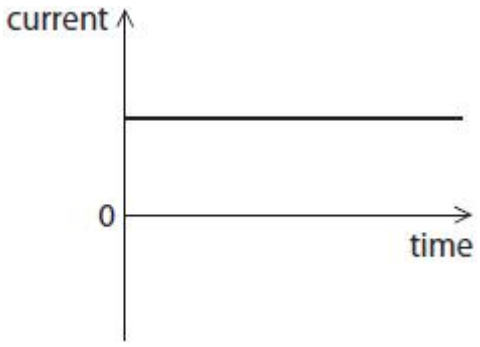

Figure 17

Explain why a transformer will not work with the input current as shown in Figure 17.

(2)

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.....

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.....

(ii) A transformer has 30 turns on the primary coil and 150 turns on the secondary coil.

A potential difference of 25 V is applied across the primary coil.
Calculate the potential difference across the secondary coil.
Use an equation selected from the list of equations at the end of this paper.

(3)

potential difference = ... V

(Total for question = 5 marks)

Q2.

This question is about a transformer.

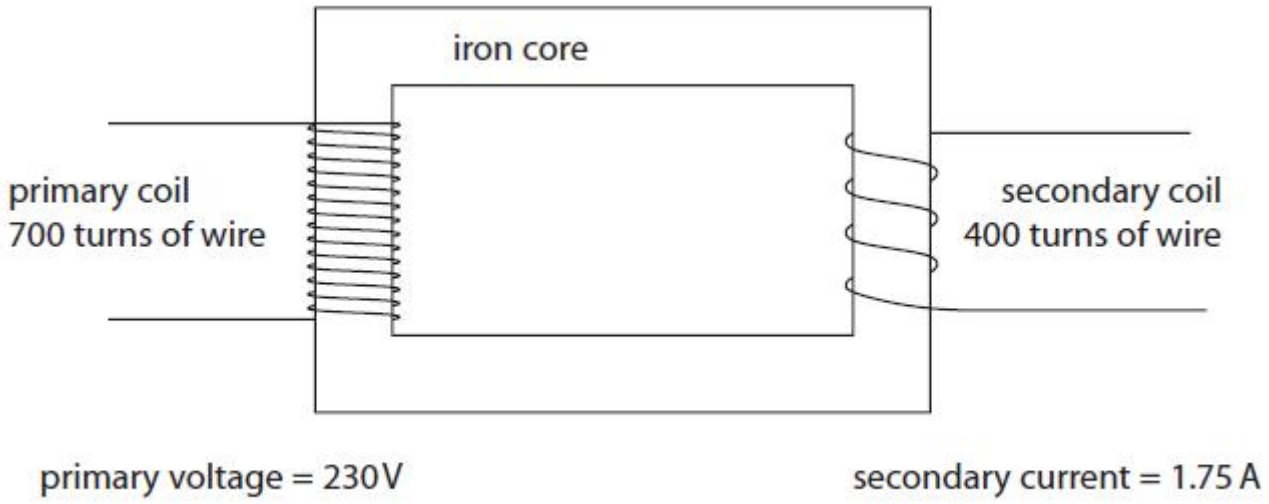

Figure 21

Calculate the current in the primary coil.

Use the information given in Figure 21 and equations selected from the list of equations at the end of this paper.

The transformer is 100% efficient.

(3)

current in the primary coil = ... A

(Total for question = 3 marks)

Q3.

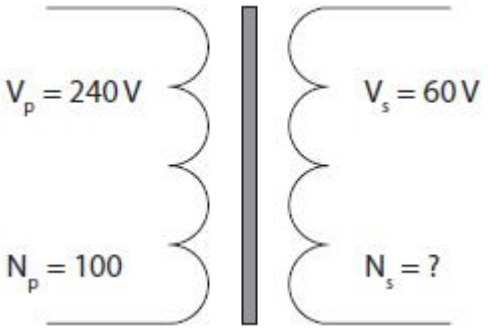

Figure 13

Figure 13 represents a transformer.
Calculate the number of turns in the secondary coil.

Use an equation selected from the list of equations at the end of this paper.

(2)

number of turns .....

(Total for question = 2 marks)

Q4.

The primary coil of a different transformer is connected to the 230 V mains supply.

The voltage across the secondary coil is 15 V.

The primary coil has 600 turns.

Calculate the number of turns on the secondary coil.

Use an equation selected from the list of equations at the end of the paper.

(2)

number of turns = ...

(Total for question = 2 marks)

Q5.

Figure 18 shows a transformer.

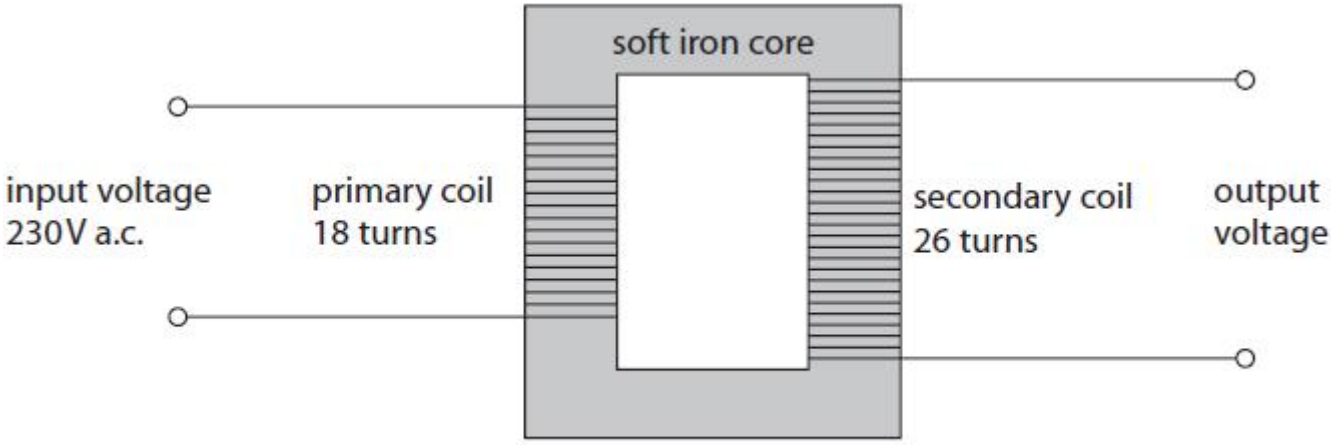

Figure 18

(i) State the purpose of the transformer shown in Figure 18.

(1)

.....

(ii) Calculate the output voltage of the secondary coil.

Use an equation selected from the list of equations at the end of this paper.

(3)

output voltage = ... V

(Total for question = 4 marks)

Q6.

Figure 25 shows a transformer.

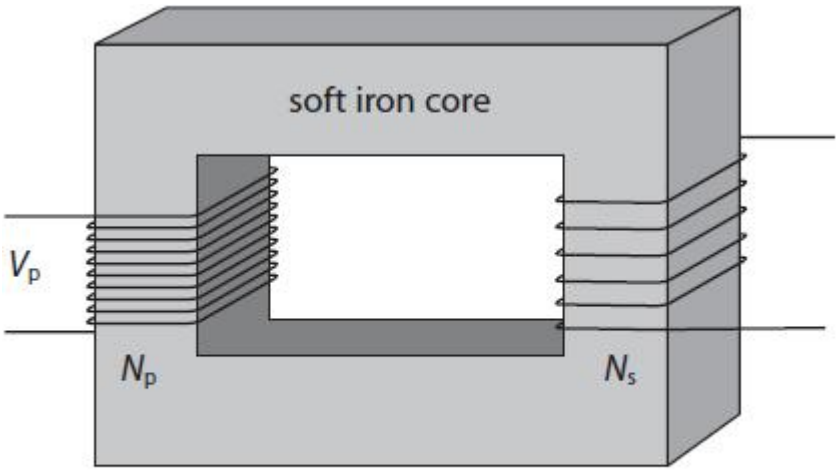

Figure 25

The number of turns on the primary coil, $N_p = 800$

The potential difference across the primary coil, $V_p = 230$ V

The number of turns on the secondary coil, $N_s = 18$

Calculate the potential difference across the secondary coil.

Use an equation selected from the list of equations at the end of the paper.

(3)

potential difference across the secondary coil = ... *V*

(Total for question = 3 marks)

Q7.

* Figure 26 shows a picture of an electrical device and a simplified drawing of the important parts.

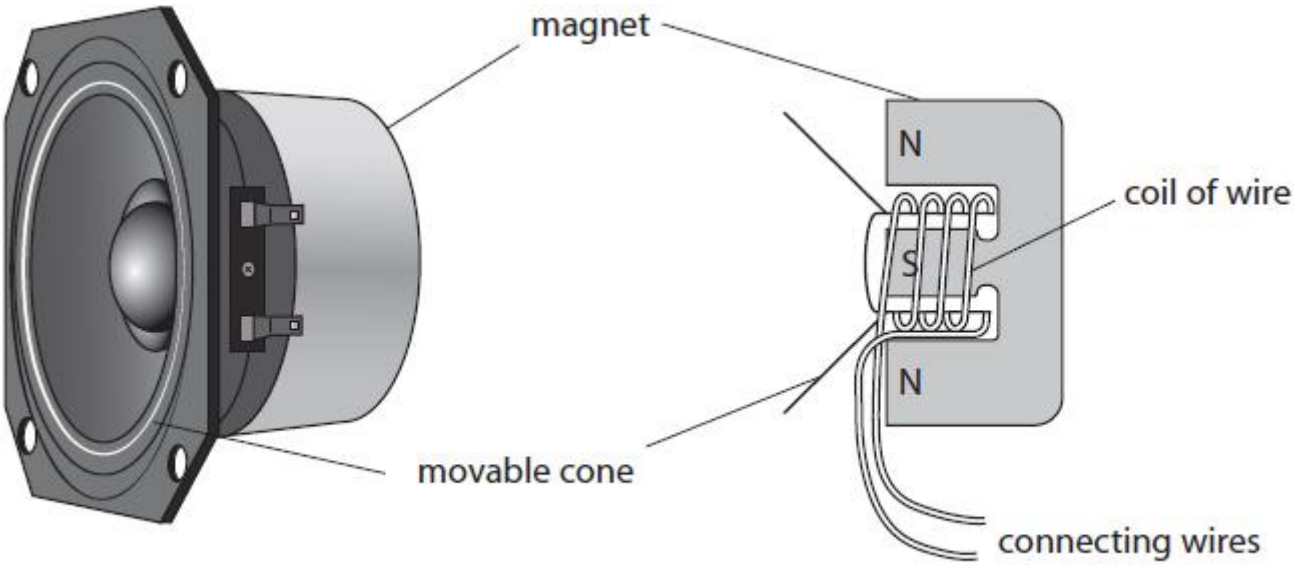

Figure 26

The device can be used as a loudspeaker or it can be used as a microphone.

Compare how the device operates when used as a loudspeaker with how the device operates when used as a microphone.

(6)

.....

.....

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.....

(Total for question = 6 marks)

Q8.

Which of these is designed to convert pressure variations in sound waves into alternating currents in

electrical circuits?

(1)

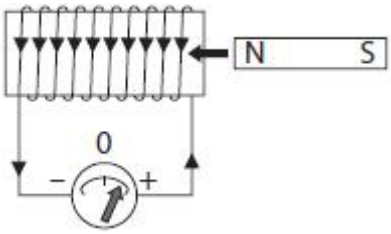
- ☐ **A** an amplifier
- ☐ **B** headphones
- ☐ **C** a loudspeaker
- ☐ **D** a microphone

(Total for question = 1 mark)

Q9.

Figure 11 shows a magnet moving into a coil.

This generates a current.



The meter deflects to the right.

Which one of these would cause a deflection to the left?

(1)

- ☐ **A** move the north pole to the left more rapidly
- ☐ **B** keep the magnet still and move the coil to the left
- ☐ **C** keep the magnet still and move the coil to the right
- ☐ **D** move the north pole in to the left and then suddenly stop it

(Total for question = 1 mark)

Q10.

There is a changing magnetic field in the core of a transformer.

(i) Describe the cause of the changing magnetic field in the core of the transformer.

(2)

.....

.....

.....

.....

(ii) A potential difference of 230 V is applied across the primary coil of a transformer.

There is a potential difference of 15 V across the secondary coil.
The primary coil has 2000 turns.
Calculate the number of turns in the secondary coil.
Use an equation selected from the list of equations at the end of this paper.

(3)

..... turns

(Total for question = 5 marks)

Q11.

What is the name of the device used to change the size of an alternating voltage?

(1)

.....

Mark Scheme

Q1.

Question Number:	Answer	Additional Guidance	Mark
(i)	an explanation linking:		(2) AO 1 1
	(p.d. / current is only induced by a) changing magnetic field (1)	alternating magnetic field	
	a changing current (is needed to create a changing magnetic field) (1)	the voltage/current (as shown) is not changing	

Question Number:	Answer	Additional Guidance	Mark
(ii)	substitution into $\frac{V_p}{V_s} = \frac{N_p}{N_s}$ (1)	substitution and rearrangement in either order	(3) AO 2 1
	$\frac{25}{V_s} = \frac{30}{150}$	$\frac{V_s}{25} = \frac{150}{30}$	
	rearrangement (1)		
	$V_s = \frac{25 \times 150}{30}$		
	evaluation (1)		
	(V_s =) 130 (V)	allow 120 or 125	
		award full marks for correct answer without working	

Q2.

Question number	Answer	Additional guidance	Mark
	selecting, rearranging and substituting (1) $(V_s = \frac{N_s}{N_p} \times V_p)$ $= \frac{400}{700} \times 230$ selecting, rearranging and substituting (1) $(I_p = \frac{V_s}{V_p} \times I_s)$ $= \frac{131(.429)}{230} \times 1.75$ evaluation (1) 1(.00) (A)	accept correct alternative calculation routes 1 mark for any voltage rounding to 130 V $\frac{130 \times 1.75}{230}$ 0.989 (A) using 130 award full marks for the correct answer without working	**(3)** **AO2.1**

Q3.

Question number	Answer	Additional guidance	Mark
	use of $\frac{V_p}{V_s} = \frac{N_p}{N_s}$ (1) transformation and evaluation (1) (N_s=) 25	award full marks for correct answer without working	**(2)**

Q4.

Question number	Answer	Additional guidance	Mark
	substitution into $\frac{Vp}{Vs} = \frac{Np}{Ns}$ (1) $\frac{230}{15} = \frac{600}{Ns}$ Rearrangement and evaluation (1) $(Ns =)\ \frac{600 \times 15}{230}$ = 39	allow substitution and rearrangement in either order accept values that round to 39 e.g. 39.13 award full marks for the correct answer without working. if no other marks scored then award 1 mark for answers of that round to 0.026 (eg 0.255) (substitution mark)	**(2)** **AO2**

Q5.

Question number	Answer	Additional guidance	Mark
(i)	{step up/increase}(output) voltage or {stepdown/ decrease}(output) current		**(1)**

Question number	Answer	Additional guidance	Mark
(ii)	substitution (1) $\frac{230}{V_s} = \frac{18}{26}$ rearrangement (1) $(V_s =) \frac{230 \times 26}{18}$ evaluation (1) 330(V)	substitution and re-arrangement in either order allow 332 (.2) (V) allow answers between322 (V) and 333 (V)where candidates have truncated an intermediate calculation 159.2 (V), 160 (V) gains 1 mark award full marks for the correct answer without working	(3)

Q6.

Question number	Answer	Additional guidance	Mark
	substitution into $\frac{V_p}{V_s} = \frac{N_p}{N_s}$ (1) $\frac{230}{V_s} = \frac{800}{18}$ rearrangment (1) $(V_s =) \frac{18 \times 230}{800}$ evaluation (1) (Vs =) 5.2 (V)	alternative method re-arrangement (1) $(V_s =) \frac{V_p \times N_s}{N_p}$ substitution (1) $(V_s =) \frac{18 \times 230}{800}$ evaluation (1) (Vs =) 5.2 (V) allow values that round to 5.2 (e.g. 5.175) award 1 mark for answers that round to 10 000 (e.g. 10 222) award full marks for correct answer without working.	**(3)** **AO2**

Q7.

Question number	Indicative content	Mark
*	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. Basic facts • Sound is vibrations (of air) • Loudspeaker emits / gives out sound • Loudspeaker is an output device • Microphone accepts / takes in sound • Microphone is an input device Principles of operation • Alternating current or (electrical) signal sent into a loudspeaker • Loudspeaker cone / coil vibrates (air) • Motor effect in loudspeaker • Microphone cone / coil is vibrated by air/sound • Alternating current / (electrical) signal is produced by microphone • Electromagnetic induction in microphone	**(6)** **AO2**
	Details of operation of loudspeaker • Alternating current or (electrical) signal in coil produces changing magnetic field • (changing) Force between magnetic field of coil and magnet. • Frequency / amplitude of sound depends on frequency / amplitude of alternating current Details of operation of microphone • Coil oscillates in a magnetic field • Generates an alternating p.d. (across the wires) / alternating current /signal • Frequency / amplitude of alternating p.d. / current depends on amplitude / frequency of the sound	

Level	Mark	Descriptor
	0	• No awardable content
Level 1	1-2	• The explanation attempts to link and apply knowledge and understanding of scientific ideas, flawed or simplistic connections made between elements in the context of the question. • Lines of reasoning are unsupported or unclear. (AO2)
Level 2	3-4	• The explanation is mostly supported through linkage and application of knowledge and understanding of scientific ideas, some logical connections made between elements in the context of the question. • Lines of reasoning mostly supported through the application of relevant evidence. (AO2)
Level 3	5-6	• The explanation is supported throughout by linkage and application of knowledge and understanding of scientific ideas, logical connections made between elements in the context of the question. • Lines of reasoning are supported by sustained application of relevant evidence. (AO2)

Level	Mark	Additional Guidance	General additional guidance – the decision within levels e.g. - At each level, as well as content, the scientific coherency of what is stated will help place the answer at the top, or the bottom, of that level.
	0	No rewardable material.	
Level 1	1–2	Additional guidance Basic facts	Possible candidate responses A loudspeaker vibrates to make sound.
Level 2	3–4	Additional guidance Principles of operation supported by basic facts. Facts may be implied by principles.	Possible candidate responses Sound vibrates the coil in the microphone. This makes a signal. OR A loudspeaker uses the motor effect to vibrate the cone and make a sound.
Level 3	5–6	Additional guidance Principles and details of operation. Principles may be implied by details. One device may be more developed than the other but both devices must be referenced.	Possible candidate responses Alternating current in the loudspeaker coil makes the coil move inside the magnet. This forces the cone to vibrate and produce sound. The microphone makes an alternating current.

Q8.

Question number	Answer	Mark
	D a microphone	**(1)**

Q9.

Question number	Answer	Mark
	B keep the magnet still and move the coil to the left	**(1)**

Q10.

Question Number	Answer	Additional guidance	Mark
(i)	A description that makes reference to an alternating /changing current (1) in the primary coil (1)	ignore references to voltage / potential difference AC accept switch on or off	**(2)**

Question Number	Answer	Additional guidance	Mark
(ii)	substitution into $\frac{Vp}{Np} = \frac{Vs}{Ns}$ (1) $\frac{230}{2000} = \frac{15}{Ns}$ rearrangement (1) $(N_S =)\ \frac{2000 \times 15}{230}$ evaluation (1) 130 (turns)	rearrangement and substitution can be in either order $\frac{230}{15} = \frac{2000}{Ns}$ using $\frac{Vp}{Vs} = \frac{Np}{Ns}$ accept answers that round to 130 or 131 (arising from rounding of intermediate evaluations) award full marks for the correct answer without working	(3)

Q11.

	Answer	Acceptable answers	Mark
	transformer	Step-up transformer Step-down transformer	(1)